

Student Performance Analysis

Hi there! If statistics and numbers make your head spin, don't worry. This report takes the academic data from our class (specifically our final exam scores, assignment scores, and study habits) and breaks it down into simple, everyday terms. No scary math formulas here—just clear, practical insights into how our class did and how we can all do better next time!

1. The Class Average & Score Spread

Think of this section as a quick snapshot of where our class stands. We looked at everyone's final exam scores (out of 100) and found some basic benchmarks:

What is it?	Our Value	What it actually means (In plain English)
The Class Average (Mean)	73.2 / 100	If we added up all student scores and shared them completely equally, everyone would get a 73.2 . This means the overall class did quite well!
The Middle Score (Median)	71.0 / 100	If we lined up every single student from the lowest score to the highest, the person standing exactly in the middle got a 71.0 . Exactly half the class scored higher than 71, and the other half scored lower.
The Most Common Score (Mode)	45.0 / 100	This is the single score that popped up most frequently in our list. A few students got exactly a 45.0 . Even though our average is high, this reminds us that some students struggled.
Score Spread (Standard Deviation)	15.3 points	This measures how "spread out" our scores are. Since it's 15.3, it tells us that most student scores landed within 15 points of our average (roughly between 58 and 88). The scores aren't identical; there is a healthy mix of different marks.

2. Where Do You Stand? (Percentiles)

Percentiles tell you what percentage of the class you scored *better* than. It's a great way to see your relative position instead of just a raw number.

Percentile Group	Score Cutoff	What this means for a student
Top 25% of the Class	86.2 or higher	If you scored an 86.2 or above , congratulations! You scored higher than 75% of your classmates. You are in the top quarter of the class.

Percentile Group	Score Cutoff	What this means for a student
The Middle 50% Range	61.5 to 86.2	This is the "main bulk" or the heart of our class. Half of all students scored somewhere in this standard range.
Bottom 25% of the Class	Below 61.5	If a score was below 61.5 , it fell in the lower quarter of the class. This highlights who might need some extra help or tutoring.
The Extremes	45.0 to 98.0	The absolute lowest score in our class was a 45.0 , and the highest score was a fantastic 98.0 .

3. Connection: What Habits Actually Drive High Scores?

We compared students' daily habits with their final exam scores to see what actually helps you pass. We measured this using a simple connection scale from **0** (no connection at all) to **1** (perfect, guaranteed connection):

- **Doing Assignments (Score: 0.99) → Extremely Strong Connection!**

This is almost a perfect 1.0! It tells us that students who did their homework and got good assignment marks almost **always** got high scores in the final exam. There is almost no exception.

- **Study Hours (Score: 0.99) → Extremely Strong Connection!**

Putting in the hours works. Students who sat down and studied consistently achieved much higher final marks than those who didn't.

- **Class Attendance (Score: 0.96) → Very Strong Connection!**

Just showing up to class makes a massive difference. High attendance correlates directly with a safe passing score.

 The Golden Takeaway:

If you want to ace this class, don't overcomplicate it! The data proves that **submitting your assignments on time** and **showing up to class** are the single best things you can do to guarantee an awesome grade.

4. What Are My Chances? (Probability)

Based on our class history, here are the real-world chances for a student:

- **Passing Chance (Scoring 50 or above): 95%**

Great news! 19 out of every 20 students passed the exam comfortably.

- **High Achiever Chance (Scoring 80 or above): 40%**

About 2 out of every 5 students achieved an 'A' or high 'B' grade. It is highly achievable if you put in the effort!

- **High Study Engagement (Studying > 5 hours/week): 35%**

Currently, only about 1 in 3 students studies more than 5 hours a week. If we can increase this number, our class average will jump even higher!

5. Checking for "Weird" Scores (Outliers)

Sometimes, a student might score a 0 or a 100 because of some extreme, unusual circumstance. We checked our data mathematically to see if we had any such "extreme weird scores" (called outliers):

- **Our Safe Range:** Any score between **24.38** and **123.38** is considered normal for our class.

- **The Verdict:** Since our lowest score was 45 and our highest was 98, every single student fit safely inside the normal range. **No weird outliers were found!** This means everyone's score represents their true effort.

6. Three Simple Steps to Improve Your Grades

If you are a student looking to boost your performance, follow this evidence-backed checklist:

1. **Never Skip an Assignment:** Since homework has a 0.99 connection to exam success, treat every assignment like a practice run for the real exam.
 2. **Show Up to Every Lecture:** Aim for over 85% attendance. Being physically present helps you absorb information effortlessly.
 3. **Set a Consistent Weekly Study Time:** Even studying just 1 hour a day (5 hours a week) puts you in the top tier of active learners!
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